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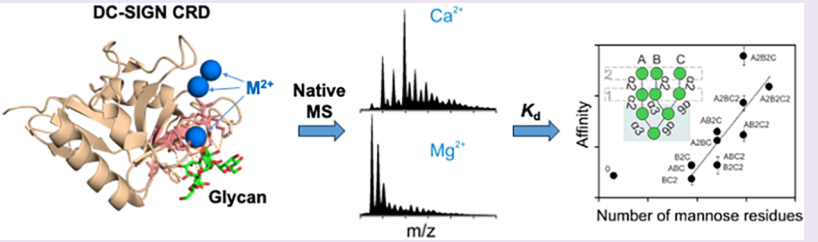
Defining Glycan Recognition by DC-SIGN Using Native Mass Spectrometry

Duong T. Bui,* Elena N. Kitova, Lara K. Mahal, and John S. Klassen*

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ABSTRACT: C-type lectins (CTLs), many of which bind carbohydrates (glycans) in a Ca^{2+}-dependent manner, are an important family of glycan-binding proteins that play essential roles in immune recognition and regulation. Among them, dendritic cell−specific ICAM-3-grabbing nonintegrin (DC-SIGN) is notable for its ability to recognize a broad range of self- and pathogen-derived glycans. Despite extensive study, the glycan binding specificity of DC-SIGN remains incompletely understood, in part because accurately quantifying intrinsic lectin−glycan binding affinities across diverse glycan repertoires remains challenging. Here, we employ native mass spectrometry (nMS), a label- and immobilization-free approach well suited for probing weak lectin−glycan interactions, to characterize the glycan-binding preferences of the carbohydrate recognition domain (CRD) of DC-SIGN. Native MS, applied directly or via a catch-and-release (CaR) strategy, enabled quantitative measurements and high-throughput screening under physiologically relevant, nonvolatile salt conditions. We further examined the binding of several divalent metal ions to the DC-SIGN CRD and their influence on glycan recognition. Although divalent cations other than Ca^{2+} bind the CRD with comparable affinity, they exert only modest effects on ligand binding. Screening a defined glycan library, including oligomannoses, human milk oligosaccharides, Lewis and blood group antigens, and ganglioside oligosaccharides, revealed a preference for oligomannoses and Lewis antigens, with affinity positively correlated with the number of mannose or fucose residues. Finally, application of CaR-nMS to a natural *N*-glycan library identified fucosylated *N*-glycans as DC-SIGN ligands. Together, these results define the glycan specificity of DC-SIGN and establish nMS as a platform for comprehensive interactome profiling of CTLs.

INTRODUCTION

Glycans—complex carbohydrates—play essential roles in molecular recognition through their interactions with glycan-binding proteins (GBPs).[1] In the immune system, glycans are recognized by lectins (nonenzymatic, nonantibody GBPs) that modulate the immune response.[2−4] Among these, C-type lectins (CTLs) represent a large family of soluble and membrane-associated proteins characterized by conserved carbohydrate-recognition domains (CRDs).[5] Many CTLs bind glycans in a Ca^{2+}-mediated manner.[5] A prominent example is the dendritic cell-specific ICAM-3−grabbing nonintegrin (DC-SIGN, CD209), which is expressed on the surface of antigen-presenting dendritic cells and certain macrophages.[5−7] DC-SIGN recognizes both self- and nonself-glycan ligands in a Ca^{2+}-dependent fashion.[8,9] Its primary ligands include fucosylated glycans, such as the Lewis blood group antigens, and oligomannose glycans commonly found in pathogens, such as HIV, Ebola virus, *Mycobacterium tuberculosis*, and *Helicobacter pylori*.[7−14] Through these interactions, DC-SIGN regulates both innate and adaptive immune responses.[7,8] The lectin can also be exploited by pathogens—for example, HIV which hijacks DC-SIGN to facilitate T-cell infection and enhance viral transmission.[15] These features have motivated the development of glycan-based therapeutics targeting DC-SIGN.[3,16,17] In addition to the canonical Ca^{2+}-dependent glycan-binding site, a recent study has revealed that DC-SIGN glycan recognition is subject to allosteric regulation through a cryptic pocket that modulates glycan affinity, highlighting the dynamic nature of the receptor's CRD.[18,19]

Despite extensive investigation, the precise glycan-binding specificity of DC-SIGN remains elusive, with conflicting findings persisting in the literature.[10−14] For example, in the case of oligomannose glycans, some studies report that the affinity

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strengthens with the number of mannose residues,[10,14] whereas others identified Man_6 or Man_7 as the optimal ligands.[11,12] Other studies have identified some fucosylated structures, such as blood group antigens, Lewis type antigens and fucosylated human milk oligosaccharides, as comparable or even better ligands for DC-SIGN.[9,14,20] Such discrepancies likely arise, in part, because monovalent DC-SIGN−glycan interactions are weak, often with a dissociation constant (K_d) in the mM range. Such weak binding approaches or exceeds the detection limits of most conventional quantitative binding assays.[14,21] Moreover, the glycan binding properties of DC-SIGN, as well as other immune lectins and GBPs, are typically profiled using glycan microarrays.[22,23] While highly effective for discovery, microarrays are not quantitative and screening requires washing steps that can obscure weak interactions and lead to false negatives. Additionally, chemical modification and surface immobilization of glycans can alter their presentation and binding properties in nonuniform ways.[24−27] Thus, there remains a pressing need for complementary analytical approaches capable of capturing weak interactions and providing a more comprehensive view of the glycan interactome of immune lectins, including DC SIGN.

As an alternative to microarray-based assays, native mass spectrometry (nMS) screening has emerged as a powerful label- and immobilization-free approach for characterizing the stoichiometry and thermodynamics of lectin−glycan complexes.[28,29] The assay is normally performed using nanoflow electrospray ionization (nanoESI) implemented under native-like solution conditions and instrumental parameters that tend to preserve noncovalent complexes in the gas phase. Direct detection of proteins and their protein−ligand complexes enables the quantification of the binding stoichiometry and affinity.[30−32] A particular advantage of nMS is the ability to detect and quantify multiple equilibria simultaneously, which enables nMS-based screening of glycan libraries.[33−35] nMS-based high throughput screening is commonly performed using the catch-and-release (CaR) approach, whereby the target GBP is used as the bait to fish out ligands from mixtures of glycans.[33,34,36] The GBP−glycan complexes are then isolated and dissociated by collision-induced dissociation to "release" the caught ligands. When performed using equimolar glycan concentrations, the relative signal intensity of the released ligands generally reflects the relative affinity and can be used for affinity ranking. Owing to these features, CaR-nMS has recently been applied to define the specificity of several lectins.[34,36,37] However, it has not yet been demonstrated for metal dependent CTLs such as DC-SIGN due to the requirement for high concentrations of nonvolatile salts. As a result, the reliability of CaR-nMS for screening CTLs is unclear and warrants examination.

Here, we apply nMS to map the glycan interactome of the DC-SIGN CRD in the presence of physiologically relevant concentrations of Ca^{2+} and establish whether divalent metals other than Ca^{2+} influence its glycan binding properties. Although Ca^{2+} is required for canonical ligand binding of DC-SIGN, it is unclear whether other physiologically relevant divalent metal ions, such as Mg^{2+}, associate with the DC-SIGN CRD or modulate glycan recognition. We first examined interactions between the DC-SIGN CRD and divalent metal ions, and then mapped DC-SIGN−glycan interactions in the presence of these metals using defined glycans representing different structural classes. These experiments confirmed that DC-SIGN binds both oligomannose and fucosylated structures, and that the presence of alternative metal ions does not measurably affect binding. While defined glycan libraries enable systematic interrogation of binding, their limited structural diversity necessitates the use of natural libraries to expand the range of glycans that can be screened. This requirement motivated our evaluation of CaR-nMS as a platform for glycan screening under the elevated Ca^{2+} conditions required for DC-SIGN function, which have not previously been demonstrated. Our results establish the feasibility of applying CaR-nMS to Ca^{2+}-dependent lectins, such as DC-SIGN, and provide a workflow that can be extended to other CTLs.

MATERIALS AND METHODS

Proteins

Bovine ubiquitin (Ubq, MW 8565 Da), lysozyme from chicken egg white (Lyz, MW 14 310 Da) and Enolase from baker's yeast (*Saccharomyces cerevisiae*, MW 93.5 kDa) were purchased from Sigma-Aldrich Canada (Oakville, Canada). The CRD of DC-SIGN (residues 250−404, MW 17, 794 Da) was prepared as described elsewhere.[38] The Ubq and Lyz stock solutions were prepared by dialysing against aqueous ammonium acetate (200 mM, pH 7.4) four times using an Amicon 0.5 mL microconcentrator with molecular weight cutoff (MWCO) of 3 kDa. The DC-SIGN CRD was dialyzed in ammonium acetate (200 mM, pH 7.4) with EDTA four times (4×) prior to the final round (6×) of dialysis in ammonium acetate (200 mM, pH 7.4) using Amicon 0.5 mL microconcentrator with MWCO of 10 kDa.

Oligosaccharides

The defined oligosaccharides used in this study (G1-G33) were purchased from Omicron Biochemicals Inc. (South Bend, IN), Omicron Biochemicals Inc. (South Bend, IN) and Chembind (Atlanta, GA). Their structures, MWs and corresponding vendors are given in Table S1. The *N*-glycans G34 and G35 were purchased from Chemily Glycoscience (Atlanta, GA, USA). A stock solution of each glycan (1 mM) was prepared in ultrafiltered Milli-Q water (EMD Millipore, Billerica, MA) and stored at −20 °C until used. Natural *N*-glycan library was produced from SARS-CoV-2 receptor binding domain and analyzed as described previously.[39] In short, the glycans were released from RBD using PNGase F (New England Biolab, Massachusetts, United States) at 37 °C for 24 h. The released glycans were purified from the protein using an Amicon 0.5 mL microconcentrator with a MW cutoff of 10 kDa. The flow-through containing glycans was lyophilized and reconstituted in mass spectrometry-grade water and stored at −20 °C until used. Approximately 5 μL of the solution was used for glycan analysis as described in the Supporting Information.

Salts

Acetate salts of calcium, nickel, manganese, magnesium, and zinc were purchased from common vendors and used without purification. Stock solutions of 10 mM of each salt were prepared in ammonium acetate (200 mM, pH 7.4).

Mass Spectrometry

Direct nMS binding measurements were performed in positive ion mode using a Q Exactive Orbitrap mass spectrometer and CaR-nMS screening experiments were carried out in negative ion mode using a Q Exactive Ultrahigh Mass Range (UHMR) Orbitrap mass spectrometer. Both instruments (Thermo Fisher Scientific, Bremen, Germany) were equipped with modified nanoESI sources. Standard uncoated (Catalogue number: ESIU-001) and submicron uncoated (ESIU-002) nanoESI emitters (AB Glycomics, Edmonton, Canada) were used to perform nanoESI. A platinum wire was inserted into the open end of the emitter, making contact with the solution, and a voltage of +0.6 kV (positive ion mode) or −0.6 kV (negative ion mode) was applied. For the direct binding measurements, the capillary temperature was 180 °C, the S-lens RF level was 100, the maximum inject time was 200 ms, and every 10 microscans were combined into a single scan. For the CaR-nMS experiments, the capillary temperature was 150 °C, the S-lens level was 100, the automatic gain control target was 5×10^6, and the

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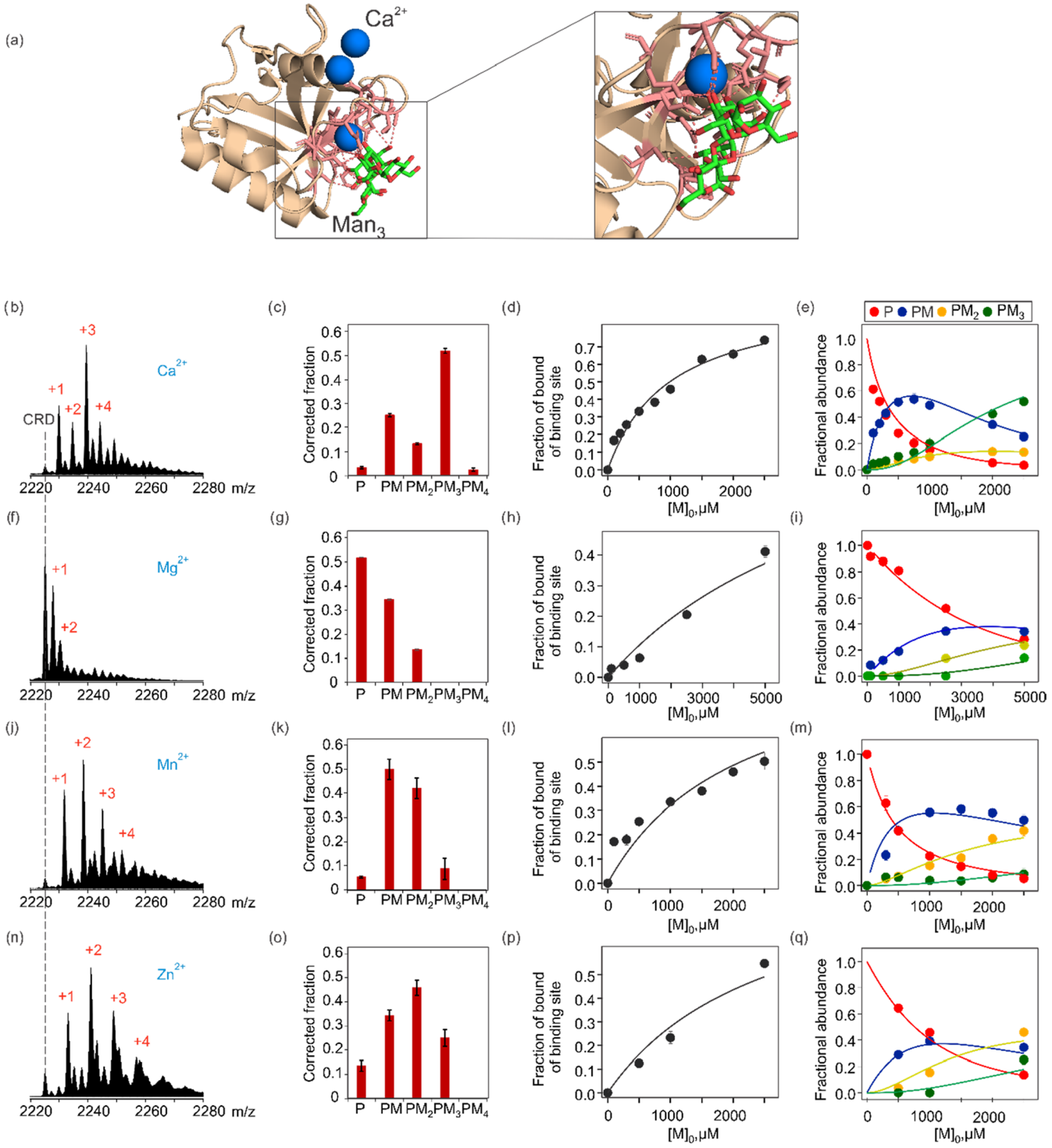


Figure 1. (a) Crystal structure of DC-SIGN CRD complexed with three $Ca^{2+}$and the glycan ligand Man_3 (PDB 2IT5). (b, f, j, n) Representative nMS spectra at charge state +8 acquired in positive ion mode for aqueous solutions (200 mM, pH 7.4) of DC-SIGN CRD (3 μM) and acetate salts of (b) Ca^{2+}, (f) Mg^{2+}, (j) Mn^{2+} and (n) Zn^{2+} (2.5 mM). (c, g, k, o) Corrected fraction of free and metal ion bound DC-SIGN CRD. Plots of concentration-dependent fraction bound of binding site and fractional abundances of free (≡P) and metal ion bound CRD (≡PM, PM_2 and PM_3) versus initial metal ion concentration ($[M]_0$) for (d, e) Ca^{2+}, (h,i) Mg^{2+}, (l, m) Mn^{2+} and (p, q) Zn^{2+}. The solid curves are the theoretical values expected for three equivalent but dependent binding sites calculated using parameters in Table 1. Error bars correspond to one standard deviation. Figures (b−e) were adapted from previous work.[21]

maximum injection time was 200 ms. Data acquisition and preprocessing were performed using the Xcalibur software (version 4.1); ion abundances were extracted using in-house software (SWARM).[40]

nMS Affinity Measurements

For a 1:1 protein−ligand (PL) interaction (eq 1), the dissociation constant (K_d) can be calculated from the concentration ratio of L-bound-to-free P (R, eq 2) at equilibrium and the initial P and L concentrations ($[P]_0$ and $[L]_0$, respectively) (eq 3). The underlying

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assumption in such measurements is that the abundance (Ab) ratio of L-bound-to-free P (R_{app}) gas phase ions reflects the concentration ratio (R) in solution[28,29]

$$P + L \rightleftharpoons PL \tag{1}$$

$$R = \frac{[PL]}{[P]} = \frac{Ab(PL)}{Ab(P)} = R_{app} \tag{2}$$

$$K_d = \frac{[L]_0 - \frac{R}{R+1}[P]_0}{R} \tag{3}$$

This assumption is typically satisfied when the MW of the ligand is less than 5% of the MW of the protein.[28,41] In principle, K_d can be calculated at a single concentration of P and L. However, for low affinity interactions it is generally advisible to use a titration approach, wherein $[P]_0$ is fixed and $[L]_0$ is systematically varied. In this case, K_d can be determined by fitting eq 4 to a plot of fraction bound protein ($R/(R+1) \equiv F$) versus $[L]_0$

$$F = \frac{[P]_0 + [L]_0 + K_d - \sqrt{(K_d - [L]_0 + [P]_0)^2 + 4K_d[L]_0}}{2[P]_0} \tag{4}$$

The DC-SIGN CRD possesses three Ca^{2+} binding sites and it is assumed that there are three binding sites for other divalent M^{2+} ions.[9,10,21,42] Stepwise M^{2+} binding can be described by eqs 5a−5c

$$P + M^{2+} \rightleftharpoons PM^{2+} \tag{5a}$$

$$PM^{2+} + M^{2+} \rightleftharpoons P(M^{2+})_2 \tag{5b}$$

$$P(M^{2+})_2 + M^{2+} \rightleftharpoons P(M^{2+})_3 \tag{5c}$$

The corresponding equations (eqs 6a and 6b) of mass balance are

$$[M^{2+}]_0 = [M^{2+}] + [PM^{2+}] + 2[P(M^{2+})_2] + 3[P(M^{2+})_3] \tag{6a}$$

$$[P]_0 = [P] + [PM^{2+}] + [P(M^{2+})_2] + [P(M^{2+})_3] \tag{6b}$$

where $[M^{2+}]_0$ is the initial concentration of the metal ion. The apparent K_d ($K_{d,i}$) for sequential addition of M^{2+} (where the i = 1−3) can be expressed by eqs 7a−7c

$$K_{d,1} = \frac{[P][M^{2+}]}{[PM^{2+}]} = \frac{[M^{2+}]_0 - 3Fs[P]_0}{R_1} \tag{7a}$$

$$K_{d,2} = \frac{[PM^{2+}][M^{2+}]}{[P(M^{2+})_2]} = \frac{[M^{2+}]_0 - 3Fs[P]_0}{R_2/R_1} \tag{7b}$$

$$K_{d,3} = \frac{[P(M^{2+})_2][M^{2+}]}{[P(M^{2+})_3]} = \frac{[M^{2+}]_0 - 3Fs[P]_0}{R_3/R_2} \tag{7c}$$

where R_1, R_2 and R_3 ($\equiv R_i$) are the abundance ratios of the M^{2+}-bound protein ($P(M^{2+})_i$) to free P measured by nMS, eq 8

$$R_i = \frac{Ab(P(M^{2+})_i}{Ab(P)} = \frac{[P(M^{2+})_i]}{[P]} \tag{8}$$

and Fs is described by eq 9

$$Fs = \frac{iR_i}{3(1 + R_i)} \tag{9}$$

If all three M^{2+} binding sites are equivalent and independent, with a common intrinsic (per binding site) K_d ($K_{d,intrin}$), $K_{d,i}$ will be related to $K_{d,intrin}$ by statistical coefficients as shown in eq 10

$$3K_{d,1} = K_{d,2} = 1/3K_{d,3} = K_{d,intrin} \tag{10}$$

If, instead, the binding sites are equivalent but dependent, such that sequential binding exhibits cooperativity (positive or negative), $K_{d,i}$ will be related to $K_{d,intrin}$ as shown in eq 11

$$3K_{d,1} = (1/\alpha)K_{d,2} = (1/3\beta)K_{d,3} = K_{d,intrin} \tag{11}$$

where α and β are cooperativity factors.

Correction for Nonspecific Metal Ion Binding

During the ESI processes, buffer and metal ions can associate with the analyte of interest forming nonspecific complexes that do not originate from the solution.[28,29] For weak interactions, higher concentrations of ligand are required to have detectable complex signal. These conditions can lead to a large fraction of nonspecific complexes. There are different ways of reducing or accounting for the nonspecific binding fraction as described elsewhere.[28,29] One attractive approach is to use emitters with smaller (submicron) diameters.[21,28,29] Submicron emitters can effectively reduce nonspecific protein-glycan complexes at glycan concentration up to 500 μM.[21,28,29] However, metal ions have high surface activity and the use of submicron tips cannot completely eliminate nonspecific interactions at high salt concentrations. In such cases, the extent of nonspecific metal binding can be corrected for using the reference protein method.[43,44] In this method, a noninteracting protein (P_{ref}), ideally of similar size as the target protein, is added to the solution. The underlying assumption is that the distributions of M^{2+} bound nonspecifically to P_{ref} and P are statistical and, therefore, identical.[43] To perform the correction, the fractional abundance of P_{ref} bound nonspecifically to q molecules of M^{2+} (f_q, where $q \geq 0$) was calculated using eq 12

$$f_q = \frac{Ab(P_{ref}(M^{2+})_q)}{\sum_q Ab(P_{ref}(M^{2+})_q)} \tag{12}$$

The true (in the absence of nonspecific binding) abundances of free and metal-bound P ($Ab(P(M^{2+})_i)$, where $i \geq 0$) are related to the apparent abundances ($Ab(P(M^{2+})_i)_{app}$) and f_q as shown in eq 13

$$Ab(P(M^{2+})_i)_{app} = f_0 Ab(P(M^{2+})_i) + f_1 Ab(P(M^{2+})_{i-1}) + f_2 Ab(P(M^{2+})_{i-2}) + \ldots + f_q Ab(P(M^{2+})_{i-q}) \tag{13}$$

Rearrangement leads to the following expressions for the true abundances of P (eq 14a), $P(M^{2+})$ (eq 14b) and, more generally, $P(M^{2+})_i$ (eq 14c)

$$Ab(P) = Ab(P)_{app}/f_0 \tag{14a}$$

$$Ab(P(M^{2+})_1) = (Ab(P(M^{2+})_1)_{app} - f_1 Ab(P))/f_0 \tag{14b}$$

$$Ab(P(M^{2+})_i) = (Ab(P(M^{2+})_i)_{app} - f_1 Ab(P(M^{2+})_{i-1}) - f_2 Ab(P(M^{2+})_{i-2}) - \ldots - f_q Ab(P(M^{2+})_{i-q}))/f_0 \tag{14c}$$

RESULTS AND DISCUSSION

Affinity of DC-SIGN for Divalent Metal Ions

DC-SIGN and many other CTLs interact with glycan ligands via a Ca^{2+} coordinator.[5] The Ca^{2+} affinity of DC-SIGN CRD and the influence of Ca^{2+} on glycan binding have been studied previously.[9,21,45,46] The CRD possesses three low affinity Ca^{2+} binding sites (Figure 1a), with an apparent K_d ($K_{d,app}$) of 1.00 ± 0.08 mM at 25 °C.[21] Sequential binding to these sites is reported to exhibit both positive and negative cooperativity, with α and β coefficients of 6 and 0.25, respectively (Figure 1a−c).[21] In cells and biological fluids (plasma, interstitial fluids), there are many other divalent metal ions that may influence the glycan-binding properties of DC-SIGN (Table S2). For example, Mg^{2+} is found at mM concentrations, comparable to those of Ca^{2+}. Other divalent cations, including Cu^{2+}, Zn^{2+}, Ni^{2+}, and Mn^{2+}, occur at substantially lower concentrations but are known to associate with a wide range of proteins and to modulate the activity of

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Table 1. Divalent Metal Ion (Ca^{2+}, Mg^{2+}, Mn^{2+} and Zn^{2+}) Affinities of DC-SIGN CRD Measured in Ammonium Acetate (200 mM, pH 7.4) by nMS[a,b]

metal ions	$K_{d,aver}$ (mM)	$K_{d,intrin}$ (mM)	α	β
Ca^{2+}	1.0 ± 0.1[c]	0.84[c]	6[c]	0.25[c]
Mg^{2+}	8.4 ± 0.7	10.5	0.7	0.4
Mn^{2+}	2.1 ± 0.3	1.4	2.3	2.3
Zn^{2+}	2.6 ± 0.4	3.3	0.6	0.6

[a]The $K_{d,aver}$ values were found assuming 3 equiv and independent metal binding sites. The $K_{d,intrin}$ values were found assuming equivalent and dependent binding sites with cooperativity factors α and β. [b]Errors correspond to one standard deviation. [c]Values taken from ref 21.

numerous enzymes.[44] Despite their biological relevance, the interactions of these metal ions with DC-SIGN and their potential impact on glycan recognition have not been systematically explored.

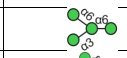
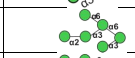
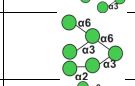
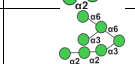
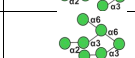
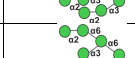
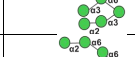
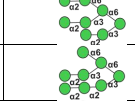
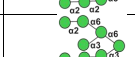
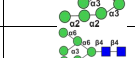
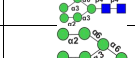
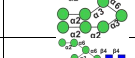
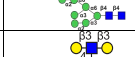
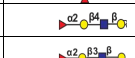
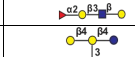
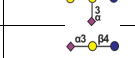
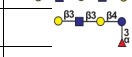
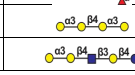
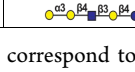
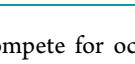
As a first step toward addressing this gap, we quantified the $K_{d,app}$ of the DC-SIGN CRD binding to Mg^{2+}, Mn^{2+} and Zn^{2+} ions using nMS and compared these to $K_{d,app}$ for Ca^{2+}.[19] Mg^{2+} and Zn^{2+} are the next most abundant metal ions in plasma (Table S2) and Mn^{2+} was included to enable a more comprehensive assessment of divalent cation binding. These measurements were performed using submicron emitters to minimize nonspecific metal ion adduction. Additionally, a P_{ref} was added to solution to correct mass spectra for any nonspecific binding that occurred during the nanoESI process. Representative mass spectra for acquired solutions of CRD and each of the four metal ions reveal markedly distinct distributions of CRD−metal ion complexes (Figures 1b,f,j,n and S1), indicating metal ion-specific binding behavior. Average K_d ($K_{d,aver}$) for each CRD−metal ion pair were obtained by fitting (eq 4) to the fraction of occupied binding sites, assuming three equivalent and independent sites. Binding is overall weak, with $K_{d,aver}$ values (Table 1) in the low mM range (1−8 mM). The CRD exhibits the weakest affinity for Mg^{2+} (8.4 ± 0.7 mM), while Mn^{2+} (2.1 ± 0.3 mM) and Zn^{2+} (2.6 ± 0.4 mM) bind with affinities comparable to Ca^{2+} (1.0 ± 0.1 mM).[19]

Although eq 4 describes the experimental data reasonably well, the assumption of equivalent and independent binding sites does not fully capture the observed binding behavior. To further interrogate this, we applied models assuming equivalent but dependent binding sites to calculate theoretical fractional abundances of free and metal ion-bound CRD, which were overlaid with the experimental data.[25] Notably, the equivalent and dependent binding-sites model provides the best description of the data for all four metal ions (Figure 1 and Table 1). For Mg^{2+}, the intrinsic K_d ($K_{d,intrin}$) is 10.5 mM, with cooperativity parameters α = 0.7 and β = 0.4. Mn^{2+} binding displays a $K_{d,intrin}$ of 1.4 mM, with weaker second and third binding events, as reflected by α and β parameters of 2.3. Zn^{2+} binds with a $K_{d,intrin}$ of 3.3 mM and exhibits positive cooperativity ($\alpha = \beta$ = 0.6, eq 11).

Effect of Divalent Metal Ions on Glycan Binding Properties of DC-SIGN CRD

Having established that the DC-SIGN CRD binds to a variety of divalent metal ions, albeit with distinct $K_{d,intrin}$ and cooperative behaviors, we next examined how metal ion identity influences glycan recognition. Because evaluating glycan binding in the presence of each metal ion individually would not reflect physiological conditions, where multiple divalent cations coexist and compete for occupancy of the CRD, we assessed CRD−glycan interactions in the presence of a physiologically relevant mixture of divalent metal ions (metal mix: Ca^{2+} 2.5 mM, Mg^{2+} 1.0 mM, Cu^{2+} 15 μM, Mn^{2+} 0.5 μM, Zn^{2+} 25 μM, Ni^{2+} 1.0 μM, Co^{2+} 1.0 μM). This approach enables evaluation of glycan

Table 2. K_d for DC-SIGN CRD and Glycans in Ammonium Acetate Solution (200 mM, pH 7.4) Containing $Ca(CH_3COO)_2$ 2.5 mM or Mixture of $Ca(CH_3COO)_2$ and $Mg(CH_3COO)_2$ 2.5 mM Each or Mixture of $Ca(CH_3COO)_2$ 2.5 mM and Acetate Salts of Divalent Metal Ions Listed in Table S2 at Their Physiological Concentrations[a]

Glycan ID	Glycan structure	K_d, mM Ca^{2+}	K_d, mM Ca^{2+} and Mg^{2+}	K_d, mM Metal mix
G1		1.26 ± 0.05	1.08 ± 0.05	1.27 ± 0.02
G2		0.79 ± 0.13		
G3		1.42 ± 0.15		
G4		0.81 ± 0.06		
G5		0.45 ± 0.07	0.40 ± 0.16	
G6		0.79 ± 0.13		
G7		0.39 ± 0.05		
G8		0.79 ± 0.14		
G9		0.41 ± 0.06		
G10		0.18 ± 0.01	0.34 ± 0.06	0.27 ± 0.01
G11		0.27 ± 0.01	0.44 ± 0.06	0.26 ± 0.01
G12		0.48 ± 0.04		
G13		0.23 ± 0.02		
G14		0.90 ± 0.11		
G15		0.93 ± 0.02	0.60 ± 0.16	0.94 ± 0.02
G16		1.06 ± 0.13		
G17		0.98 ± 0.02	0.60 ± 0.10	1.00 ± 0.03
G18		0.62 ± 0.03	1.17 ± 0.19	0.71 ± 0.02
G19		6.5 ± 0.2	6.3 ± 1.7	6.9 ± 0.3
G20		3.3 ± 0.6	4.5 ± 1.3	
G21		2.8 ± 0.5		
G22		1.1 ± 0.1		
G23		7.0 ± 1.1		
G24		4.5± 1.4		
G25		23 ± 6		
G26		7.0 ± 2.3	4.4 ± 0.7	
G27		8.7 ± 1.3	NB	
G28		8.6 ± 1.5		
G29		4.1 ± 1.3	4.3 ± 1.1	
G30		9.2 ± 2.9	2.9 ± 1.0	
G31		0.96 ± 0.03	0.73 ± 0.14	0.92 ± 0.01
G32		5.6 ± 2.6		
G33		4.4 ± 0.6		

[a]Errors correspond to one standard deviation.

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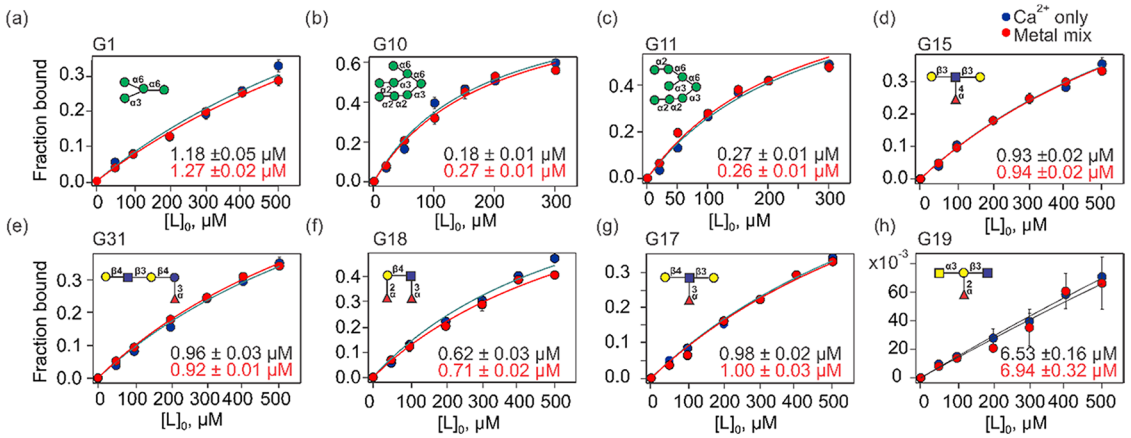


Figure 2. Plots of fraction of DC-SIGN bound by glycans (a) G1, (b) G10, (c) G11, (d) G15, (e) G31, (f) G18, (g) G17 and (h) G19 at different glycan concentrations measured by nMS using aqueous solutions (200 mM, pH 7.4) of DC-SIGN CRD (3 μM) in the presence of 2.5 mM $Ca(CH_3COO)_2$ (blue circles) or metal mixture at their physiological concentrations (red circles). The solid curves are the best fit of eq 4 to the experimental data. Error corresponds to one standard deviation.

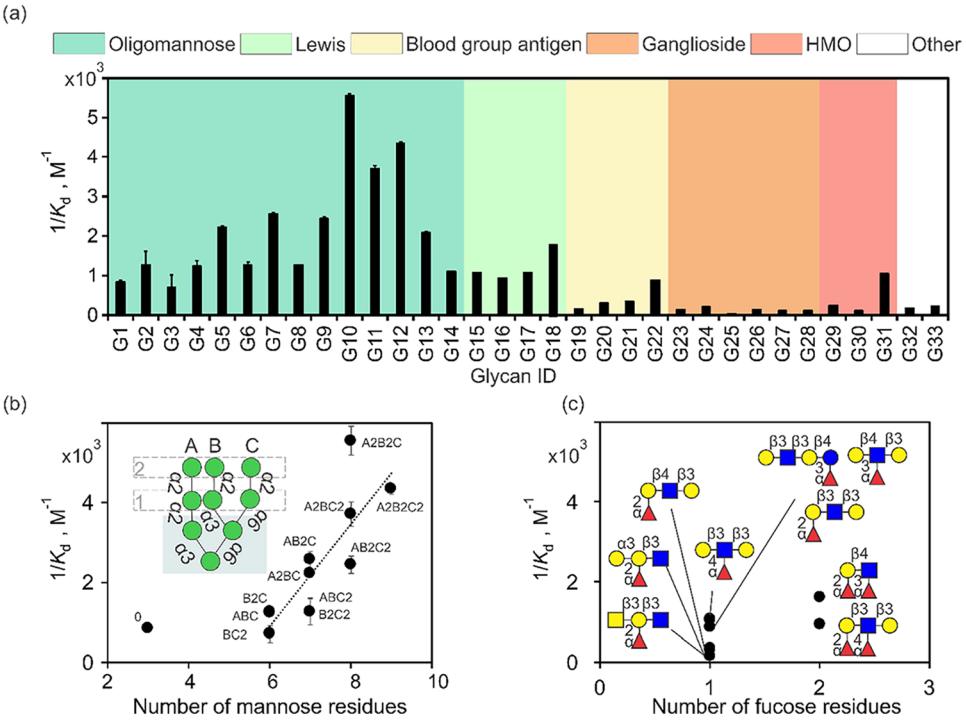


Figure 3. Interaction of DC-SIGN CRD with different glycan types (Table 2) measured by nMS. (a) $1/K_d$ values for binding of DC-SIGN CRD (3 μM) to structurally defined glycans measured in aqueous solution (200 mM, pH 7.4) in the presence of $Ca(CH_3COO)_2$ (2.5 mM). (b, c) Correlations between the number of (b) mannose residues and (c) fucose residues and the corresponding $1/K_d$. The terminal α1,2−linked mannose residues are labeled A, B, and C for the branches and 1 or 2 for the number of residues on each branch extending from the trimannosyl core structure (indicated as 0 and highlighted by the gray box). Error corresponds to one standard deviation.

binding under competitive metal-binding equilibria that more closely approximate in vivo conditions (Table S2). Binding measurements were performed in the presence of either Ca^{2+} alone (2.5 mM) or the metal mix. Affinities were determined for eight mannose- and fucose-containing glycans (G1, G10, G11, G15, G17−G19 and G31) (Table 2 and Figure 2). Additionally, to further probe the potential influence of Mg^{2+}, which exhibits measurable intrinsic affinity for the CRD and is present at millimolar concentrations in vivo, the affinities of 14 glycans were measured for solutions containing equimolar (2.5 mM) Ca^{2+} and Mg^{2+} (Table 2). In both experiments, a P_{ref} was included to correct for nonspecific metal ion and glycan adduction.[19] Representative mass spectra are shown in Figure S2.

Importantly, the presence of other metal ions (in addition to 2.5 mM Ca^{2+}) in solution is found to have no statistically significant effect on the strength of glycan binding by DC-SIGN CRD. This observation is consistent with the well-established role of Ca^{2+} as the specific metal ion required for glycan recognition by the CRD. The lack of any measurable influence from other physiologically relevant divalent metal ions indicates that binding of these ions to metal ion binding sites of the CRD does not affect ligand recognition. Consequently, the glycan binding interactions observed can be attributed specifically to

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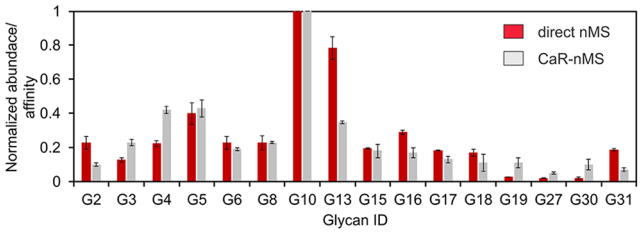


Figure 4. Comparison of the relative abundance of the released glycan ligands in CaR-nMS experiments for ammonium acetate solution (200 mM, pH 7.4) with 2.5 mM of $Ca(CH_3COO)_2$ of DC-SIGN CRD (10 μM) and glycan libraries (1 μM) and the normalized $1/K_d$ values obtained by nMS for ammonium acetate solution (200 mM, pH 7.4) with 2.5 mM of $Ca(CH_3COO)_2$ of DC-SIGN CRD (3 μM) and glycans (50−500 μM). The corresponding K_d values are listed in Table 2.

the Ca^{2+}-dependent recognition. Having established that divalent metal ions other than Ca^{2+} at physiological concentrations do not affect glycan binding, all subsequent binding measurements were performed with the DC-SIGN CRD in the presence of Ca^{2+} alone.

Glycan Specificity of DC-SIGN

We next carried out direct nMS binding measurements using a titration approach on 33 defined glycans representing five classes: oligomannoses, blood group antigens, Lewis-type antigens, human milk oligosaccharides (HMOs), ganglioside oligosaccharides, as well as two other neutral glycans (Gal4 (G32) and galili pentaose (G33)). The K_d values obtained are shown in Figure 3 (as $1/K_d$) and are listed in Table 2.

Among the glycans tested, oligomannoses exhibited the highest affinities, with G10 identified as the strongest binder. The relationship between affinity and the number mannose residues and the branching pattern is highlighted in Figure 3b. Overall, affinity increases with glycan size, in qualitative agreement with previous reports by van Die and co-workers.[14] However, the greater precision and absolute nature of the affinity measurements in the present study allow more specific glycan recognition motifs to be identified. Notably, glycan binding is influenced not only by the total number of mannose residues but also by their spatial arrangement. The presence of terminal α1,2−linked mannose residues on branches A and/or B enhances affinity, whereas an α1,2−linked mannose on branch C is less preferred. Consistent with this trend, G10, which contains terminal α1,2−linked mannose residues on both branches A and B, displays a higher affinity than G11, in which branches A and C are occupied. In contrast, G13, despite having the largest number of mannose residues, exhibits a weaker interaction, attributable to the terminal mannose on branch C, which may sterically hinder binding. The preference of DC-SIGN for glycans with α1,2−linked mannose on the A or B branches has not been previously described.

The fucosylated glycans have affinities comparable to Man_6 and Man_7. However, due to the limited number and diversity of fucosylated glycans tested, it was not possible to establish a comprehensive view of how fucosylation modulates binding. Of the glycans investigated, the difucosylated G18, which has a $Lewis^Y$ epitope, exhibited the strongest binding (Figure 3c). While DC-SIGN is known to exhibit enhanced avidity toward multivalent $Lewis^X$ motifs,[47] our data show that increased fucosylation within Lewis Y−type glycans similarly leads to a decrease in K_d (i.e., increased affinity). This suggests that increasing fucose density can enhance binding, likely through avidity effects or improved engagement with the CRD. These findings indicate that fucose content contributes positively to DC-SIGN binding, although the precise impact remains dependent on glycan presentation and epitope geometry. The available data further suggest that the fucose linkage influences affinity, with branched α2-linked fucose exhibiting lower affinity than α3- and α4-linked fucose. This observation is consistent with the findings of Arda and co-workers.[9] Notably, only weak binding was observed for glycans lacking mannose and fucose residues. In these cases, CRD binding is likely dominated by interactions with the galactose residue, as reported previously.[48,49]

Validation of CaR-nMS for Rapidly Screening Glycan Libraries against DC-SIGN CRD

The CaR-nMS assay has emerged as a powerful label-free method for screening defined and natural libraries of glycans against lectins to discover ligands and evaluate binding specifity.[34,36,37] This approach assumes comparable release efficiencies and MS response factors across all ligands in the library. However, its reliability for screening C-type lectins in the presence of high concentrations (mM) of divalent metal ions has not yet been established. With the view toward establishing CaR-nMS for profiling glycan specificities of CTLs, we used the assay to screen three defined glycan libraries (consisting of 16 of the 33 glycans in Table 2: G2, G3, G4-G6, G8, G10, G13 are oligomannoses, G15-G19 are Lewis and blood group antigens, G27 is a ganglioside oliogosaccharide, G30 andG31 are HMOs) against DC-SIGN CRD. Two glycans (G10 and G13) were added to each library for signal normalization. All solutions contained 2.5 mM of $Ca(CH_3COO)_2$.

Optimization of the method was carried out using DC-SIGN CRD and a library of ten glycans. Representative CaR-nMS data are shown in Figure S3. In previous CaR-nMS screening campaigns using defined libraries, glycan concentrations of 10 nM to 100 nM were employed.[34,37] However, in the present work, performed at high concentrations of $Ca(CH_3COO)_2$ and using submicron nanoESI emitters, higher concentrations, between 0.5 μM and 1 μM, were necessary for reliable detection of the released ligands. Negative ion mode was employed in these measurements to facilitate the detection of the released glycan ligands. As discussed previously, glycans readily form deprotonated ions in the gas phase, resulting in improved detection efficiency compared with positive ion mode.[34,37]

A range of collision energies (CEs) was tested (Figure S3c) and the optimal range was found to be 45 to 65 V. Importantly, comparison of the normalized abundance of released glycans (corresponding to the average measured at a CE of 50, 55, and 60 V) with the normalized affinity ($1/K_d$) obtained by direct nMS measurements shows good agreement overall (Figure 4).

Having established that CaR-nMS-based screening can be used to evaluate the glycan binding properties of DC-SIGN CRD, we next demonstrated its application for screening natural libraries of glycans.

Application of CaR-nMS for Screening Natural Glycans against DC-SIGN CRD

DC-SIGN is an important host factor in SARS-CoV-2 infection, with multiple studies reporting its ability to recognize the viral spike protein and serve as a potential attachment receptor.[47,48] Notably, this interaction can be disrupted by DC-SIGN−targeted glycomimetic antagonists,[48] strongly implicating glycan-mediated recognition rather than direct protein−protein contacts. Despite this, the specific viral glycans responsible for

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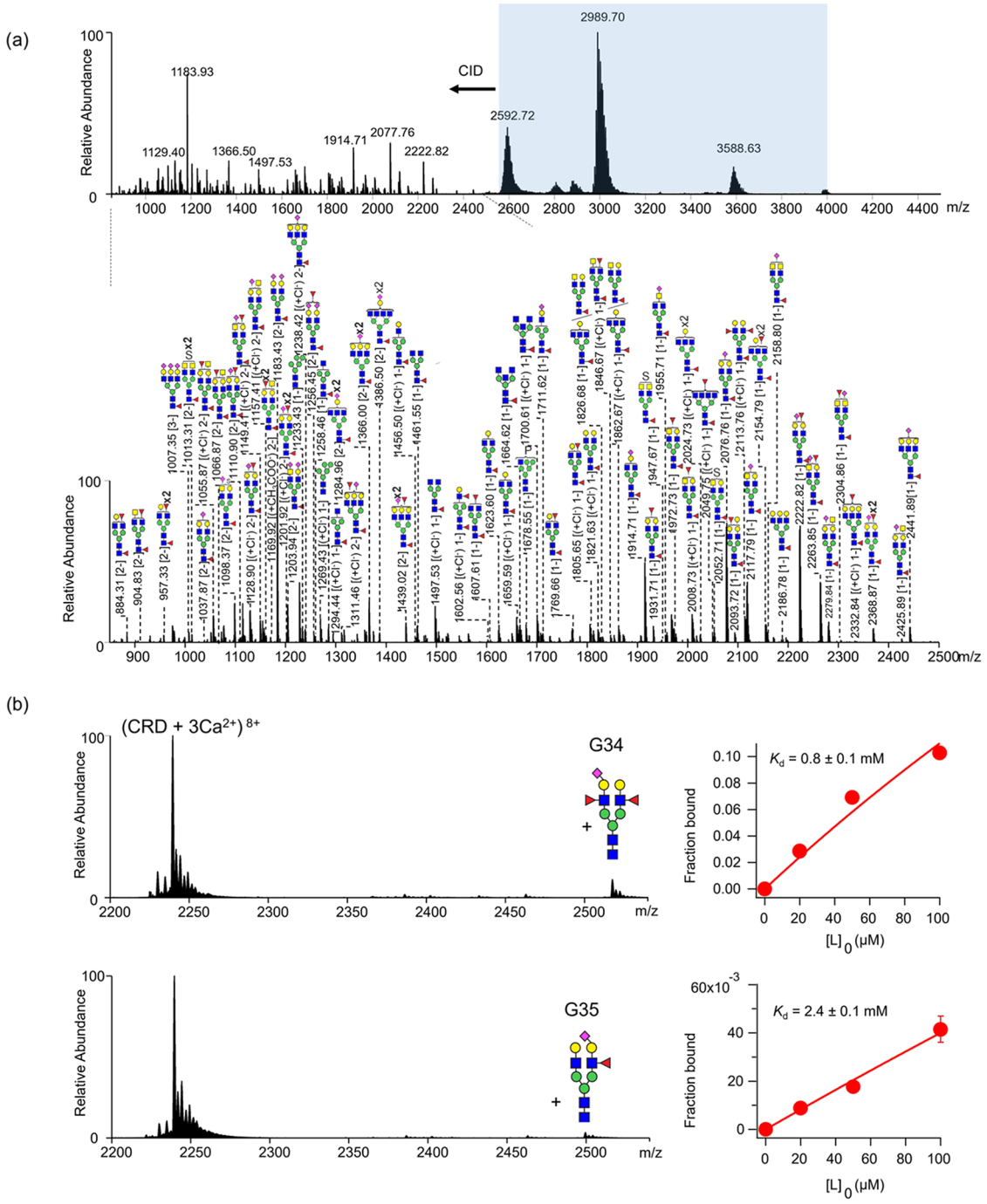


Figure 5. *N*-glycan ligand binding to DC-SIGN CRD measured by (CaR)-nMS. (a) Representative mass spectrum acquired by CaR-nMS performed in negative ion mode for an aqueous solution (200 mM ammonium acetate, pH 7.4) of DC-SIGN CRD (10 μM) and $Ca(CH_3COO)_2$ (2.5 mM) and an *N*-glycan library extracted from SARS-CoV-2 RBD. A collision energy of 60 V was used. The location of fucose on the *N*-glycans in the natural library is putative. (b) Representative mass spectra acquired by nMS performed in positive ion mode for an aqueous solution (200 mM ammonium acetate, pH 7.4) of DC-SIGN CRD (10 μM) and $Ca(CH_3COO)_2$ (2.5 mM) with purified *N*-glycan (G34 or G35, 50 μM). The region corresponding to the +8 charge state of the free and bound CRD is shown. Also shown is the concentration dependence of the fraction of CRD bound to *N*-glycan ligand (G34 or G35) measured by nMS. The solid curves are the best fit of eq 4 to the experimental data.

DC-SIGN engagement remain poorly defined. To identify glycan features contributing to DC-SIGN recognition, we employed CaR-nMS to screen a native *N*-glycan library derived from the recombinant SARS-CoV-2 receptor-binding domain (RBD). This RBD-derived library (Figure S4) reflects the structural diversity of *N*-linked glycans presented on a key viral subdomain involved in host recognition.

Using CaR-nMS, 46 MW-distinct *N*-glycan structures were detected as ligands (Figure 5a). The dominant species were mono- and difucosylated biantennary and triantennary *N*-glycans, with a subset additionally carrying sulfate or phosphate modifications. Interactions between DC-SIGN and fucosylated *N*-glycans on CHO cells have been reported previously, and Lewis antigens ($Lewis^X$ and LNDF), identified here and elsewhere as DC-SIGN ligands, constitute structural motifs

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within *N*-glycans.[50,50−52] To our knowledge, however, affinities of complex-type *N*-glycans for DC-SIGN have not been reported.

Owing to the low abundance of individual glycans in the natural library and their relatively weak binding, reliable affinity quantification using concentration-independent (COIN)-nMS was not feasible. Despite these limitations, we obtained a benchmark estimate of *N*-glycan binding by performing quantitative nMS measurements on two purified fucosylated *N*-glycans (G34 and G35) bearing one and two fucose residues, respectively. The resulting K_d values, in the low-mM range, are consistent with trends observed in the defined glycan library screen.

Notably, increased fucosylation enhanced binding to the DC-SIGN CRD, with the addition of a second fucose resulting in an approximately 3-fold increase in affinity. Together, these data provide a quantitative reference for interpreting the broader ligand screen and motivate future studies using expanded panels of purified glycans to systematically define structure−affinity relationships.

CONCLUSIONS

DC-SIGN, along with other CTLs, plays a central role in innate immune recognition and contributes to a broad range of physiological and pathological processes, including host−pathogen interactions and immune modulation. Despite extensive investigation, the glycan interactome of DC-SIGN and the molecular determinants governing its binding specificity have proven difficult to define, in part because of the challenges associated with accurately measuring intrinsic lectin−glycan binding affinities. Here, we employed nMS−based approaches to quantify the metal ion affinities of the DC-SIGN CRD, elucidate the role of divalent cations in modulating CRD−glycan interactions, and characterize glycan binding preferences.

Our results reveal that the CRD binds to a variety of divalent metal ions, although these interactions are generally weaker than those of Ca^{2+}. Moreover, glycan binding measurements performed in the presence of a mixture of divalent metal ions at their physiologically relevant concentrations failed to detect any significant enhancement in affinity relative to solutions containing only Ca^{2+}. This finding is consistent with the well-established role of Ca^{2+} as the key metal ion required for optimal glycan recognition by DC-SIGN. Affinity data acquired for defined libraries of glycans revealed that both the number and spatial arrangement of Man residues are key determinants of ligand recognition by DC-SIGN CRD. Notably, we identified a previously unreported binding motif associated with high-mannose glycans. In contrast, for fucosylated glycans, the available data did not allow unambiguous assignment of a specific binding motif, reflecting the limited structural diversity of the defined glycan library examined. To overcome this limitation, we developed a CaR-nMS workflow suitable for high-throughput glycan library screening against DC-SIGN (and other CTLs). Application of the assay to a natural library of *N*-glycans derived from the SARS-CoV-2 RBD revealed that mono- and difucosylated *N*-glycans, including several sulfated or phosphorylated species, are ligands of the DC-SIGN CRD. Quantitative nMS measurements performed on two purified fucosylated *N*-glycans indicate that these interactions are relatively weak in comparison to the oligomannoses described above, with K_d in the mM range. Ongoing efforts are directed toward applying this workflow to expanded, defined libraries, as well as natural glycan and glycopeptide libraries, to more comprehensively define the glycan specificity of DC-SIGN and other CTLs.

ASSOCIATED CONTENT

Data Availability Statement

Raw mass spectrometry data files are available upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acschembio.6c00397.

Glycan library analysis, structure and molecular weights of the defined glycans, physiological metal ion concentrations, K_d values, CaR-nMS optimization details (PDF)

AUTHOR INFORMATION

Corresponding Authors

Duong T. Bui − *Department of Chemistry, University of Alberta, Edmonton, Alberta T6G 2G2, Canada;* Present Address: Departments of Chemistry and Biology, University of Waterloo, Waterloo, Ontario N2L 3G1, Canada; orcid.org/0000-0001-9274-0759; Email: buithi@ualberta.ca

John S. Klassen − *Department of Chemistry, University of Alberta, Edmonton, Alberta T6G 2G2, Canada;* orcid.org/0000-0002-3389-7112; Phone: (780) 492-3501; Email: john.klassen@ualberta.ca

Authors

Elena N. Kitova − *Department of Chemistry, University of Alberta, Edmonton, Alberta T6G 2G2, Canada*

Lara K. Mahal − *Department of Chemistry, University of Alberta, Edmonton, Alberta T6G 2G2, Canada;* orcid.org/0000-0003-4791-8524

Complete contact information is available at: https://pubs.acs.org/10.1021/acschembio.6c00397

Author Contributions

D.T.B: Conceptualization, methodology, investigation, writing—original draft, writing—review and editing, data curation. E.N.K: Investigation, writing—original draft, writing-review and editing. L.K.M: Supervision, writing—original draft, writing—review and editing, funding acquisition. J.S.K: Supervision, conceptualization, writing—original draft, writing—review and editing, funding acquisition.

Notes

The authors declare no competing financial interest.

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